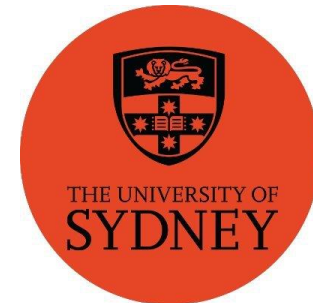


Survival Analysis of Breast Cancer

Vanessa Huang, Melissa Cheong, Keyah Loau

Mentor: Dr Carissa Chen



SUDATA
SYDNEY UNIVERSITY DATA SOCIETY

SUPPORTED BY
USU

Background

Research Motivation



1 in 7

women are at risk of being
diagnosed with breast cancer

Through our research, we hope to...



Identify strong factors of
breast cancer survival to help
future diagnosis

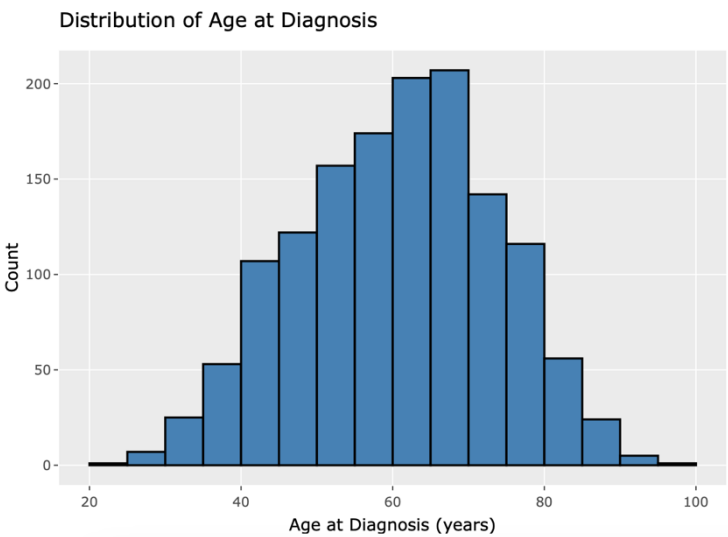


Gain deeper insights into how
survival outcomes vary cross
these key clinical and
molecular factors.

Breast cancer is the
2nd

most common cancer overall in Australia

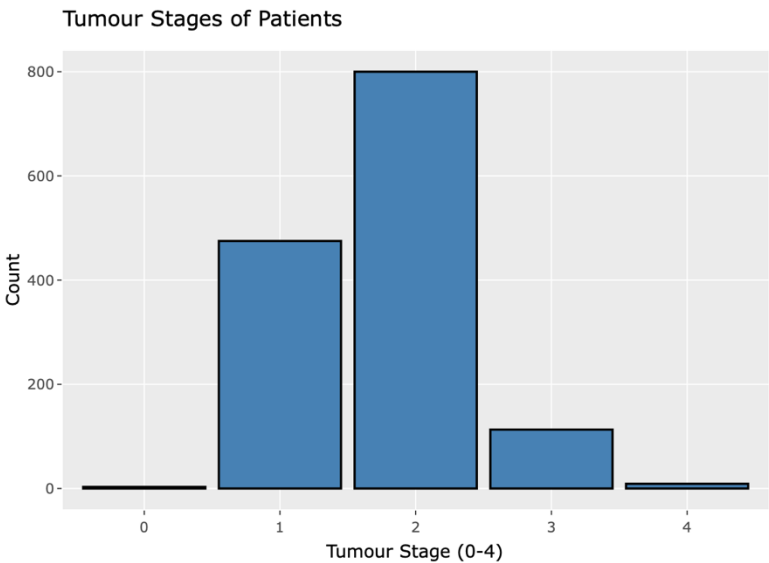
Dataset overview



1st quartile 51 years old

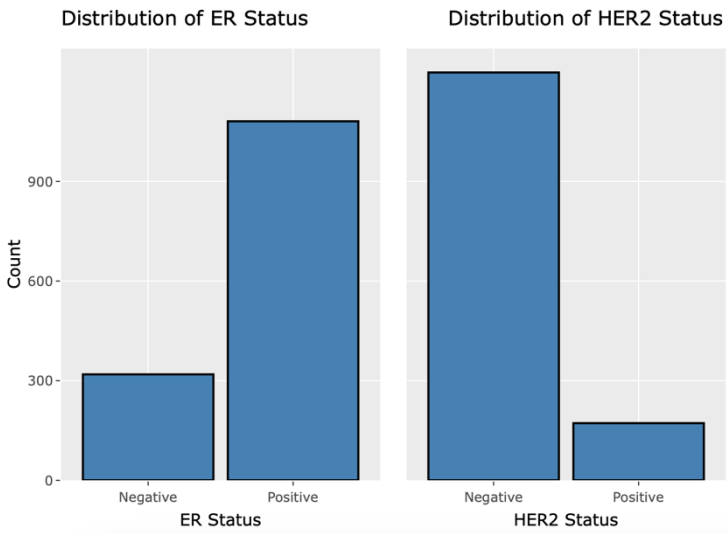
Median 60 years old

3rd quartile 70 years old



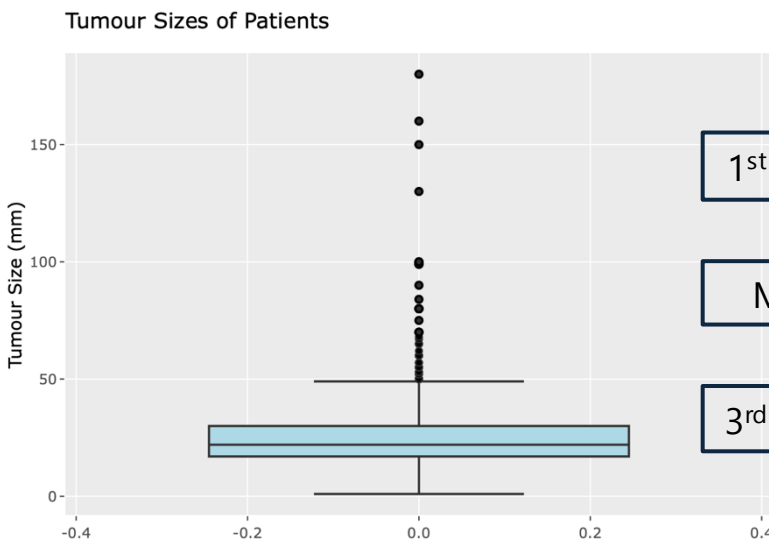
58%
of diagnosed
patients had
tumour stage 2

32%
of diagnosed
patients had
tumour stage 1



77%
of diagnosed
patients had
ER positive breast cancer

87%
of diagnosed patients had
HER2 negative breast cancer

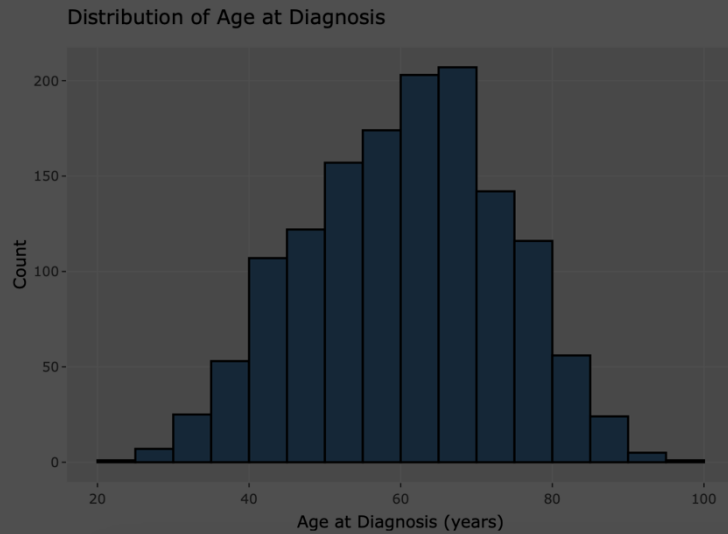


1st quartile 17.00 mm

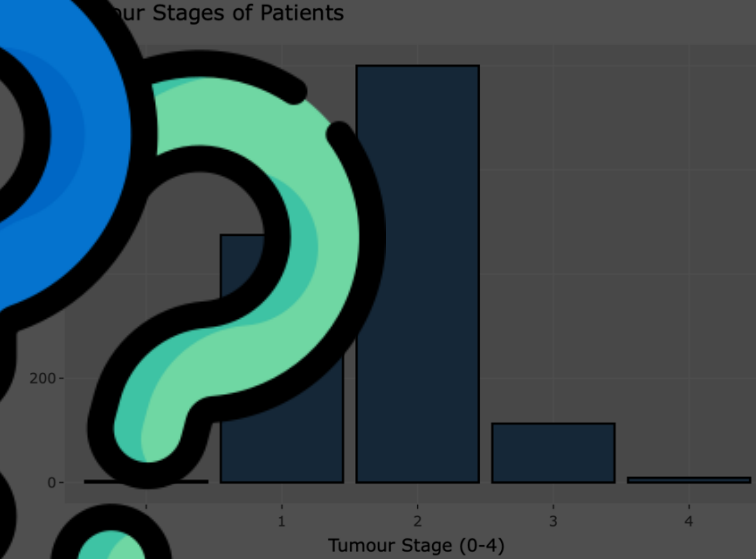
Median 22.00 mm

3rd quartile 30.00 mm

Dataset overview



1st quartile 56 years old
Median 61 years old
3rd quartile 66 years old

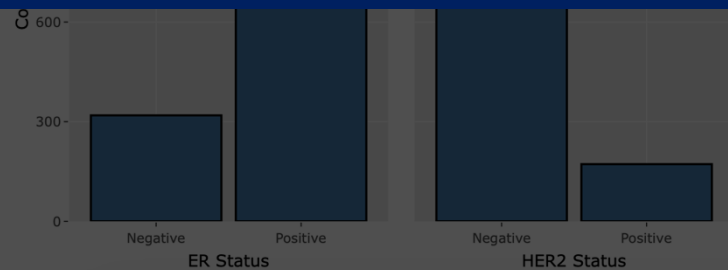


58%
of diagnosed
patients had
tumour stage 3

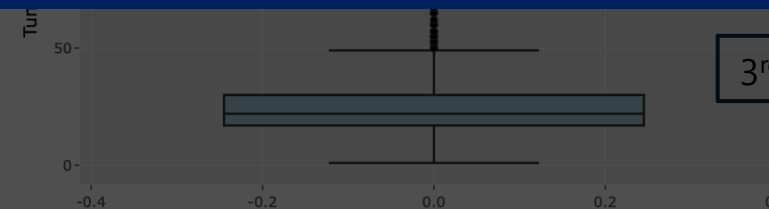
32%
of diagnosed
patients had
tumour stage 2

Research question:

To what extent do clinical attributes (age, tumour stage & size, HER2 status, ER status and number of lymph nodes) independently influence long-term survival in breast cancer patients?

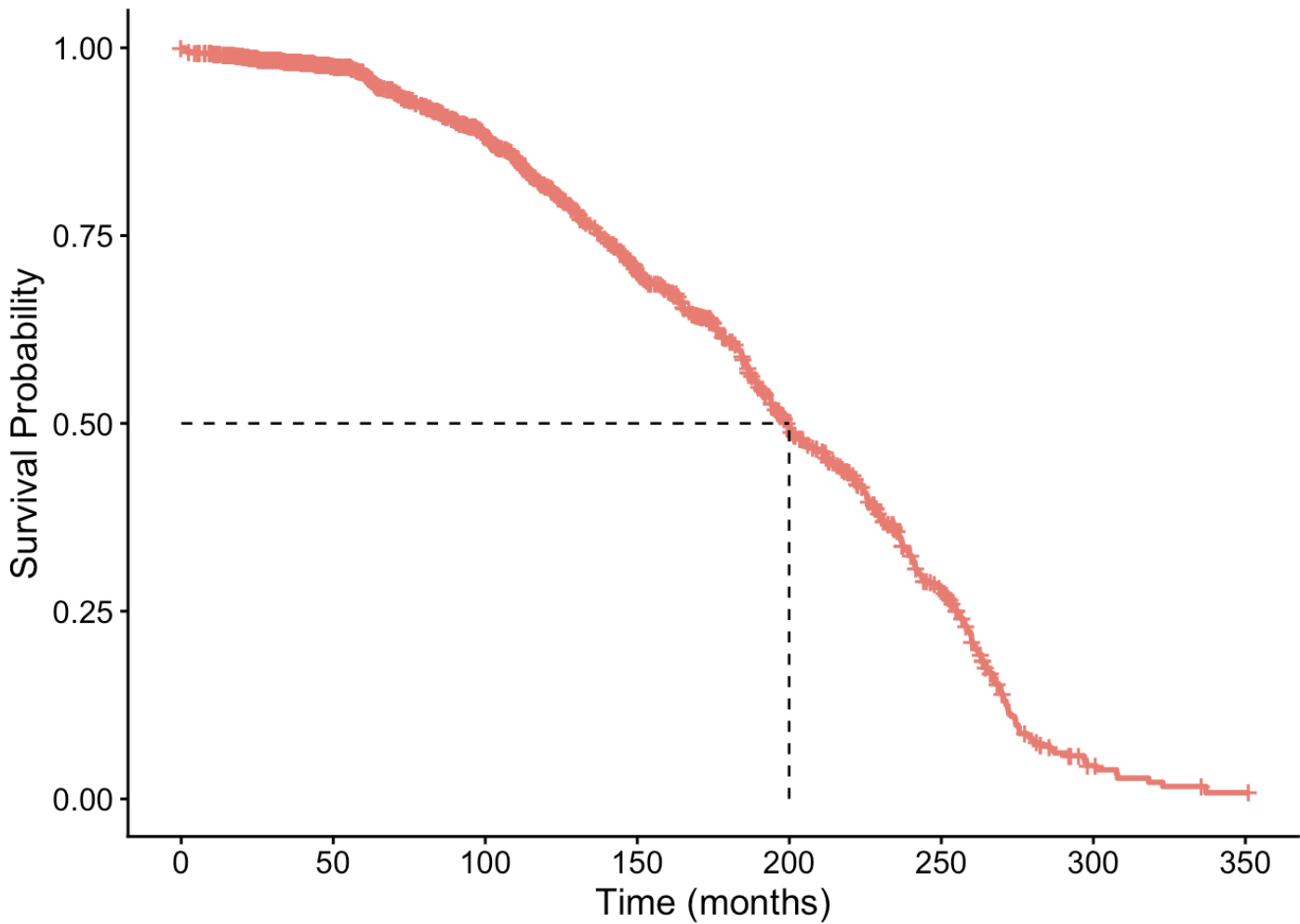


87%
of diagnosed patients had
HER2 positive breast cancer



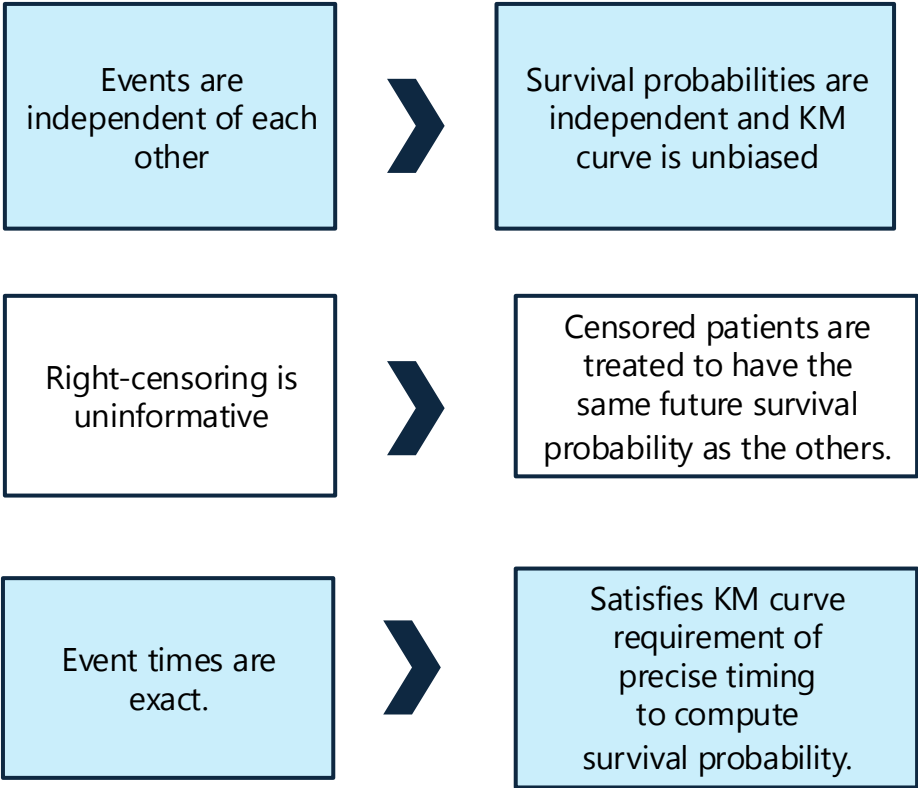
Kaplan-Meier Survival Curve Overview

Survival Probability



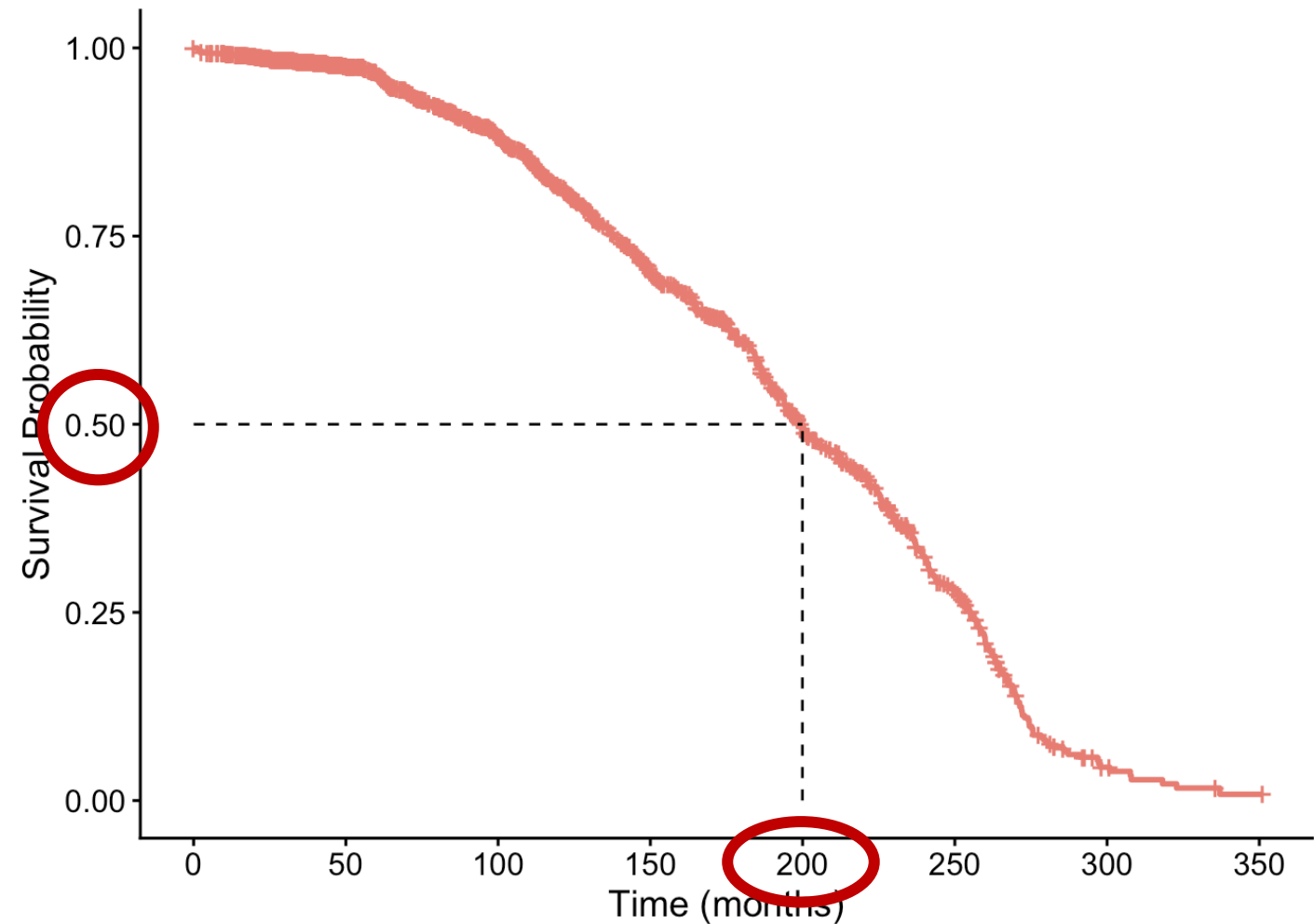
A graphical tool used in survival analysis to estimate the probability of an event happening over time.

Key assumptions made...



Kaplan-Meier Survival Curve Overview

Survival Probability



After data cleaning...

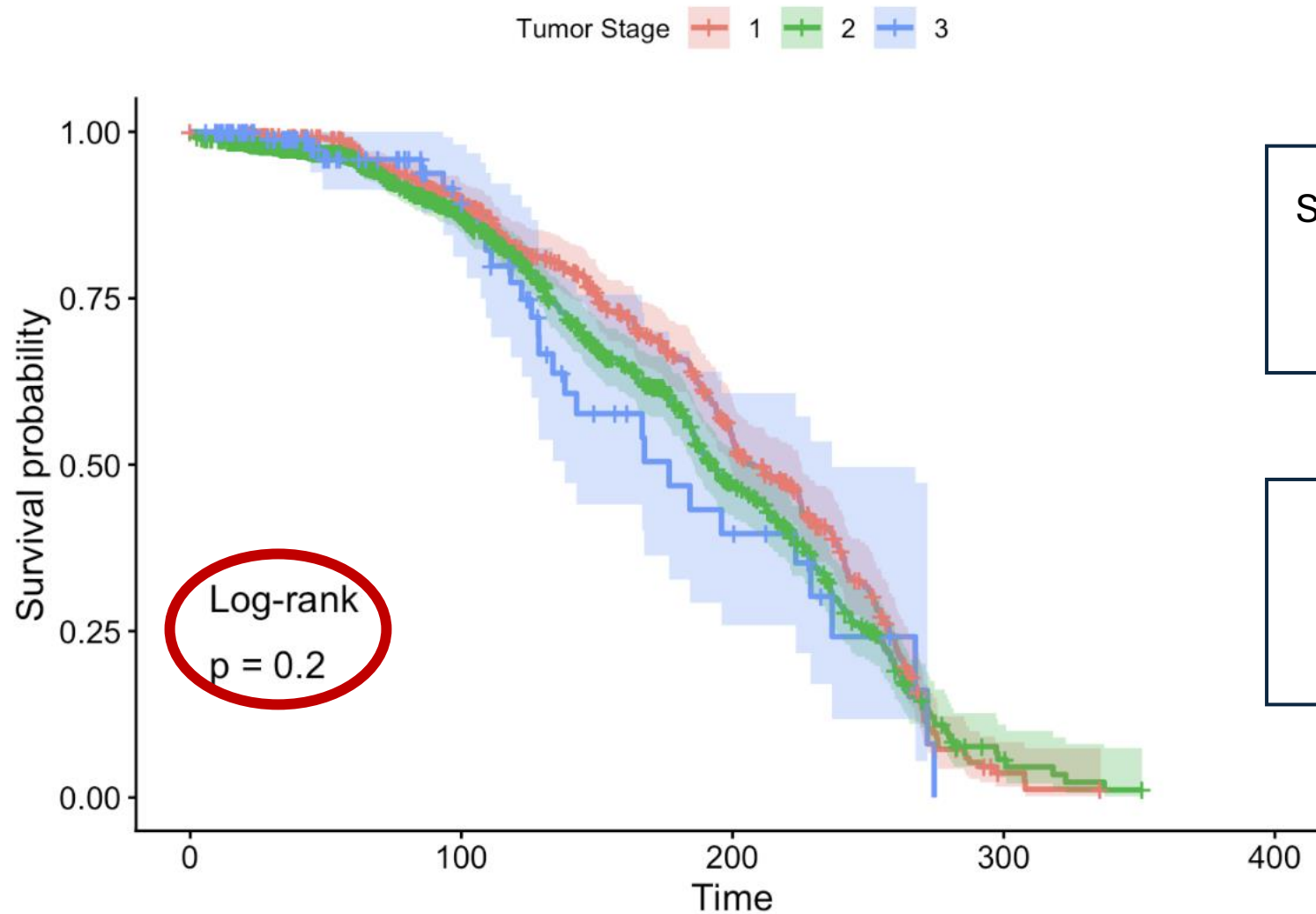
1400 patients left in the dataset	
610 (43.5%) patients experienced the event (death)	790 (56.5%) patients were censored (event was not observed)



The median **survival time** is
16 years and 8 months
(200 months)

After 16 years and 8 months
(200 months),
the **survival probability** of
patients drops below
0.5

Overview of Tumour Stage

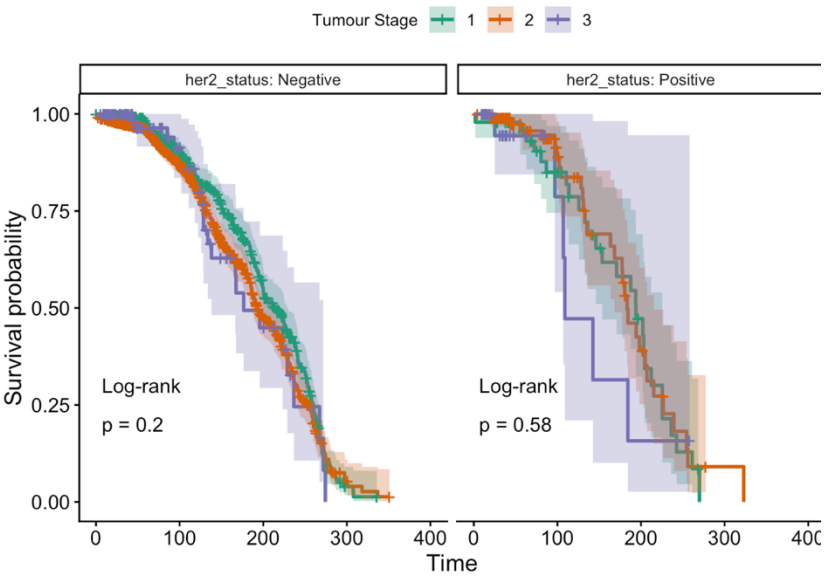


Stage 1 patients generally had the **highest survival probability**, followed by stage 2 and lastly, tumour stage 3

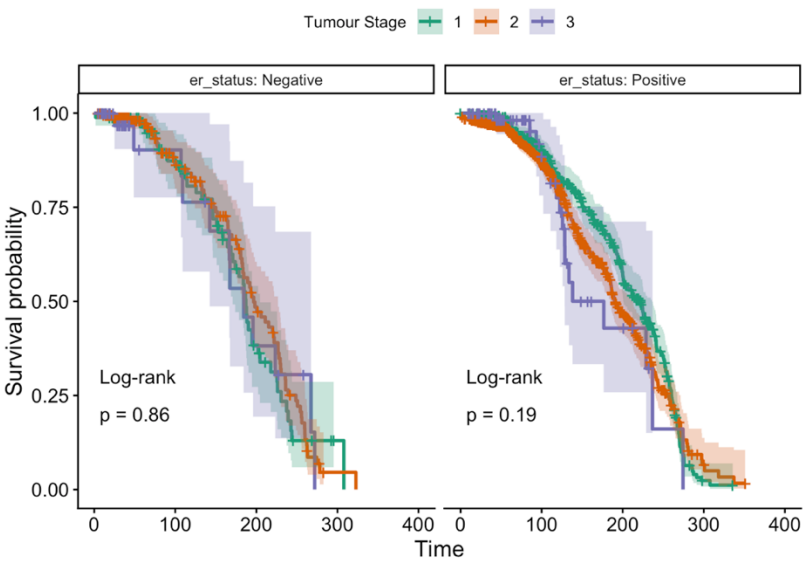
However, since $p = 0.2 > 0.05$, there is **no significant difference** in survival between tumour stages

Stratification of Tumour Stage Status by HER2 Status, ER Status and Lymph Node Size

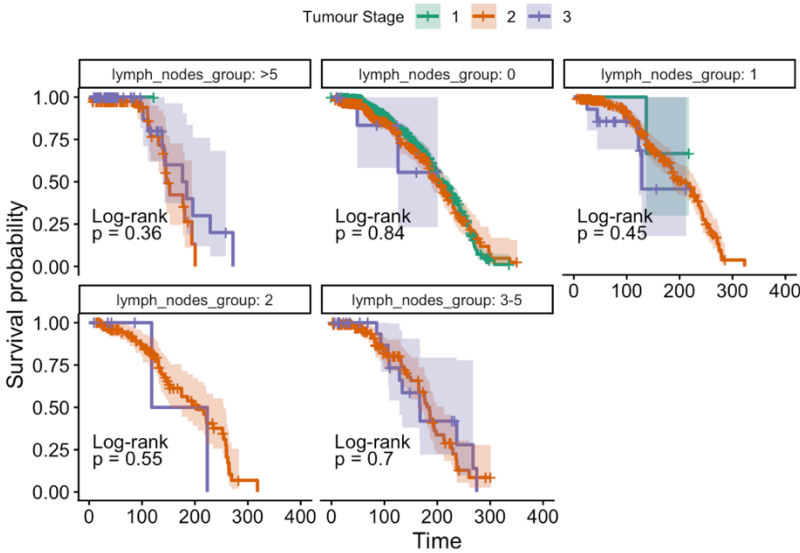
Stratified by HER2 status



Stratified by ER status

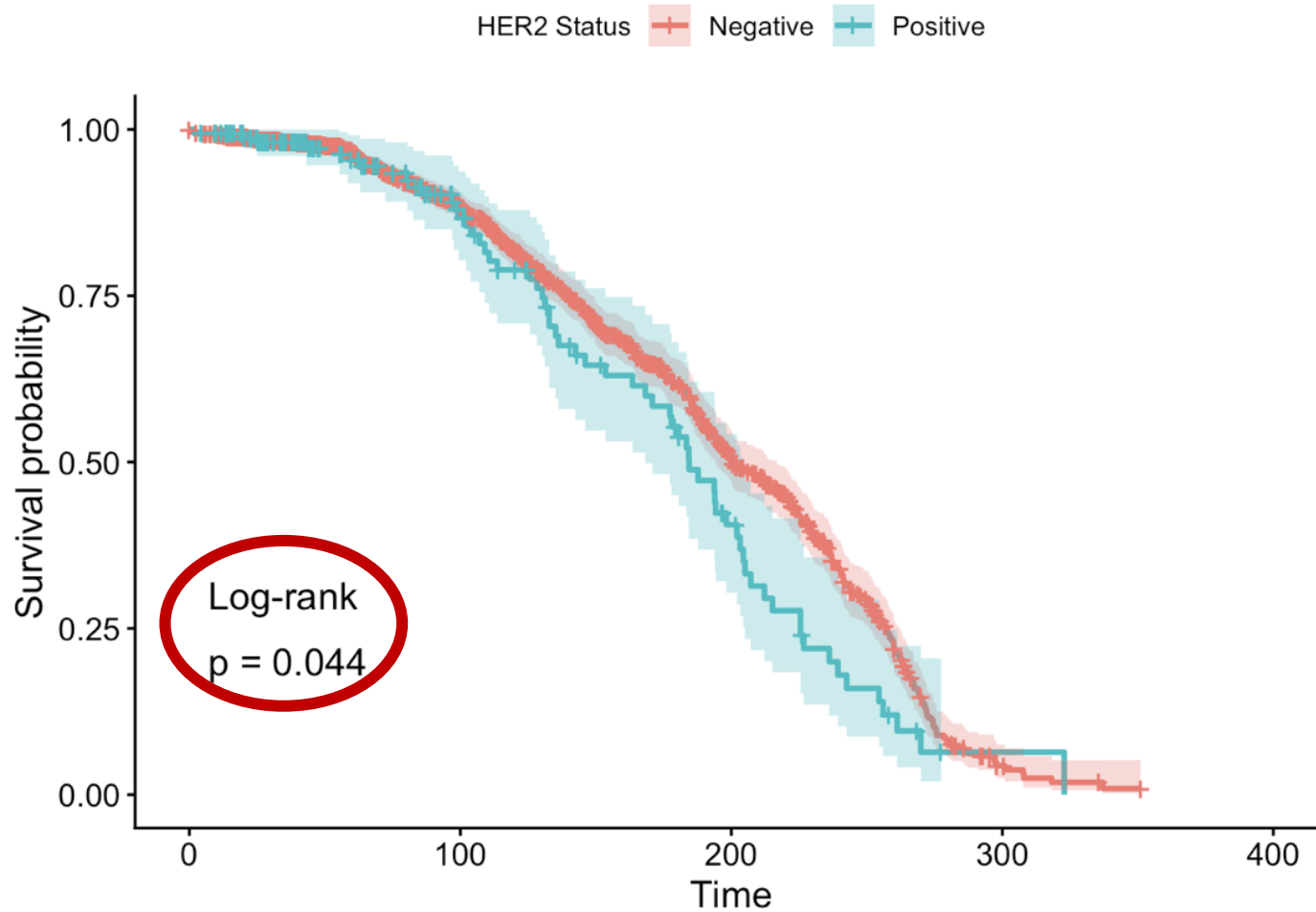


Stratified by lymph node size



Tumour stage was **not** significantly associated with survival probability when stratified by HER2 status, ER status, or lymph node size.

Overview of HER2 Status

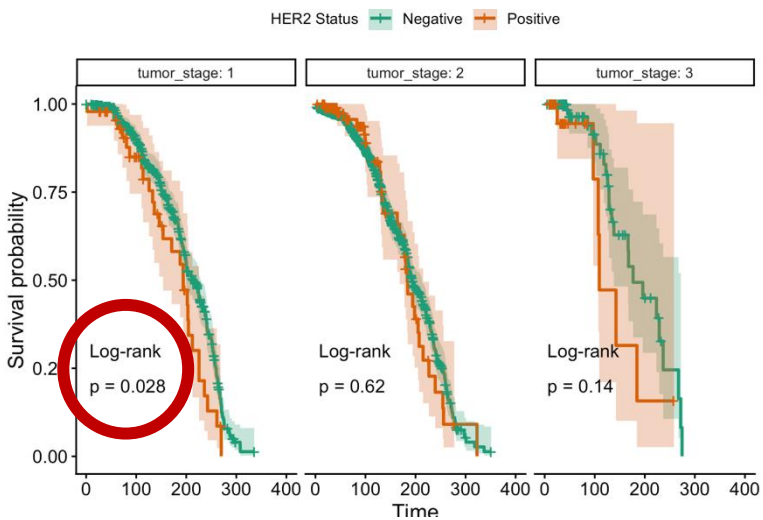


HER2 status **significantly influences** survival probability ($p = 0.044 < 0.05$)

HER2-negative patients generally had a **higher survival probability** than HER2-positive patients

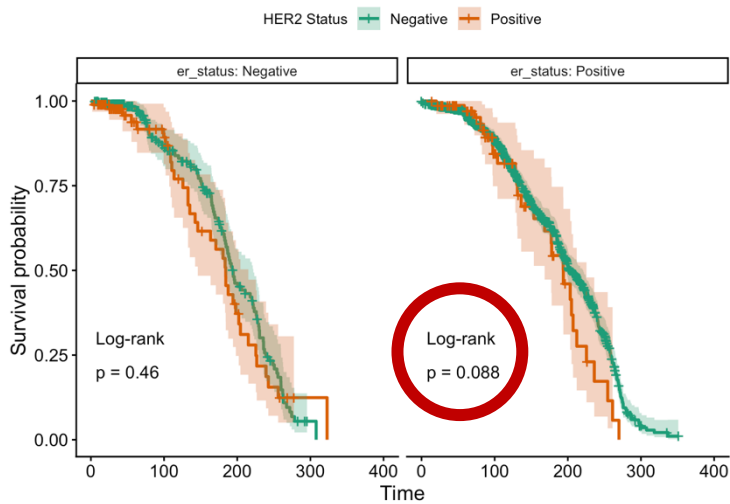
Stratification of HER2 status by Tumour stage, ER status and lymph node sizes

Stratified by Tumour stage



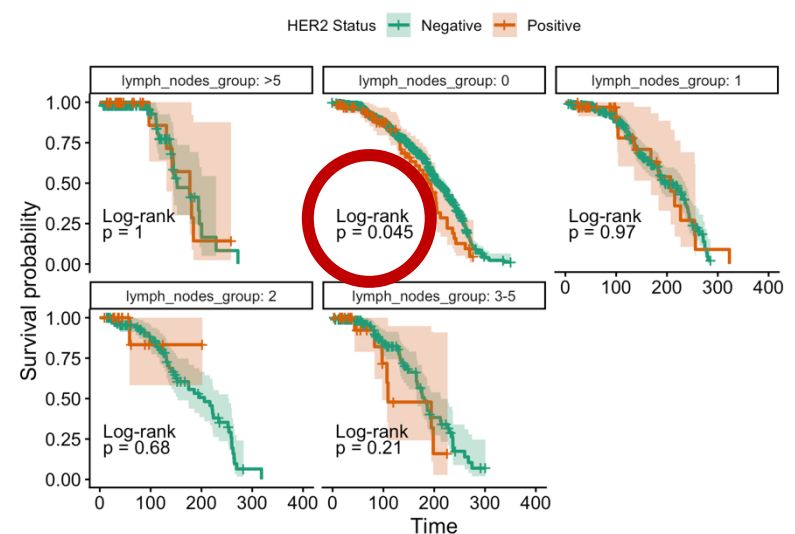
For stage 1 tumour stage patients, HER2-negative patients have a significantly higher chance of survival than HER2-positive patients. ($p = 0.028 < 0.05$)

Stratified by ER status



HER2 status has no significant effect on survival probability for ER-positive and ER-negative patients.

Stratified by lymph node size



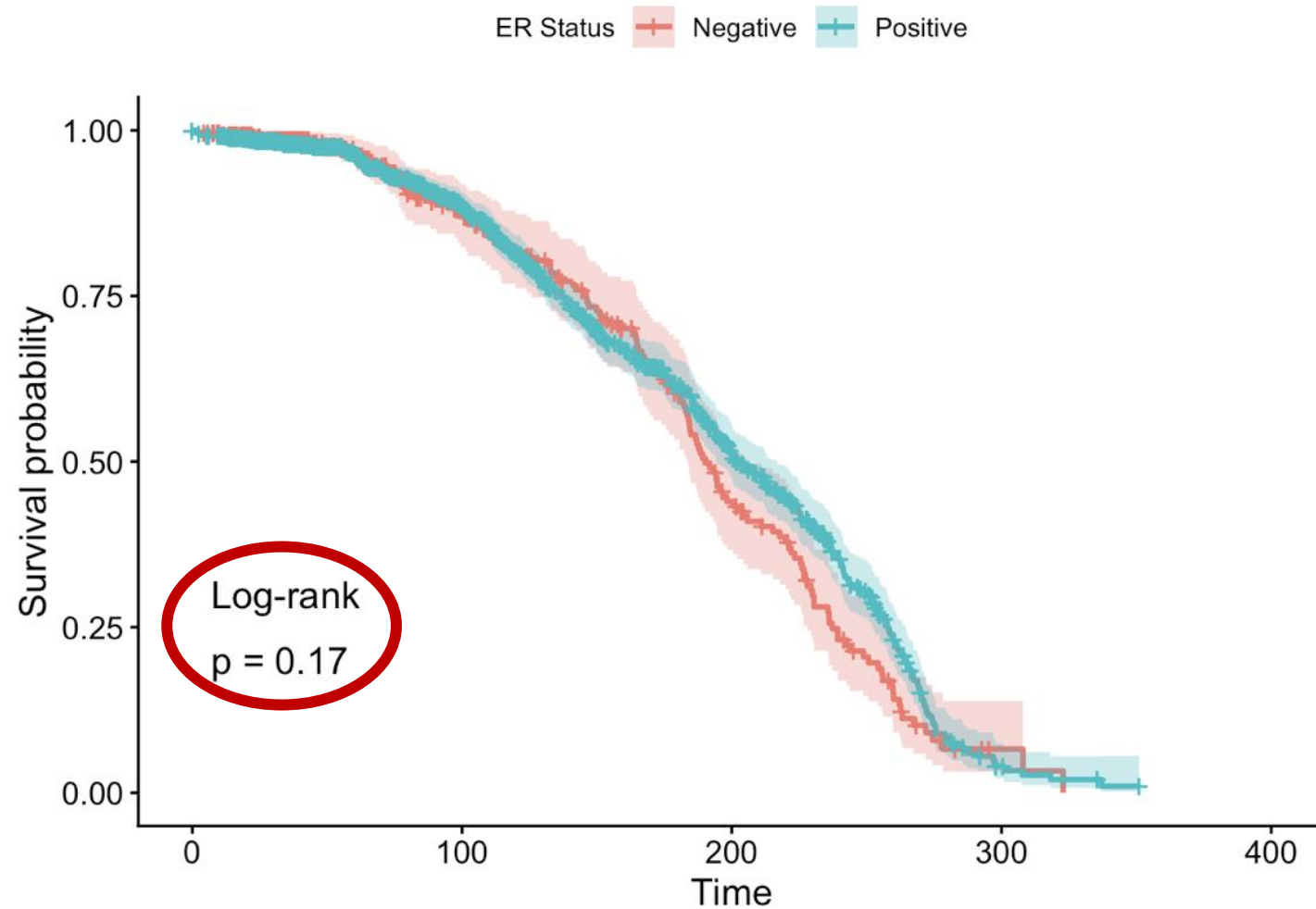
Since $p\text{-value} = 0.045 < 0.05$,

HER2 status has a significant effect on survival probability for patients with lymph node = 0.

For patients with lymph node = 0,

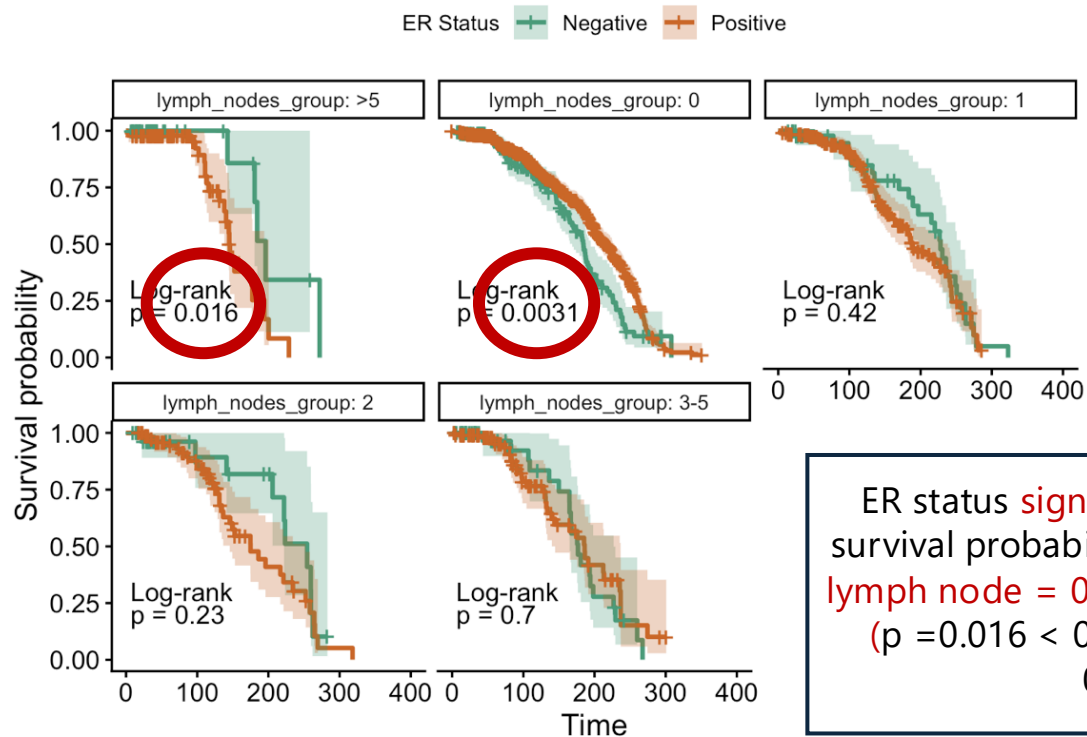
HER2-negative patients have a significantly higher chance of survival than HER2-positive patients.

Overview of ER status



Stratification of ER status by Tumour stage, HER2 status and lymph node sizes

Stratified by lymph node size



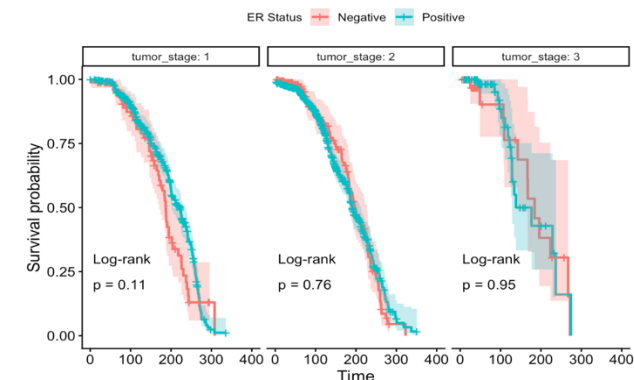
ER status **significantly influences** survival probability for patients with **lymph node = 0 and lymph node >5** ($p = 0.016 < 0.05$, $p = 0.0031 < 0.05$).

For lymph node = 0,
ER+ patients have a **significantly higher** chance of survival than ER- patients.

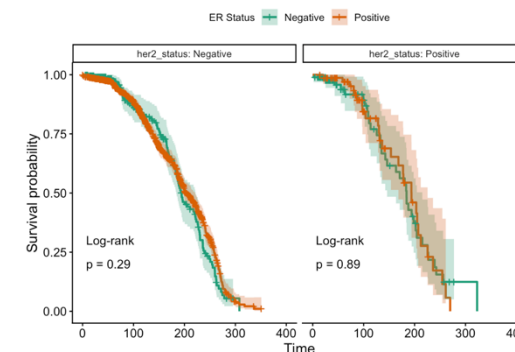
For patients with lymph node > 5,
ER- patients have a **significantly higher** chance of survival than ER+ patients.

ER status does **not significantly influence** survival probability when patients were stratified by tumour stage or by HER2 Status:

Stratified by Tumour Stage



Stratified by HER2 status



The Cox Proportional Hazards (PH Model)

We want to quantify the influence of covariates, and see which ones are significant. We do this using the Cox PH Model:

$$h(t|\mathbf{X}) = h_0(t)\exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_q X_q)$$

Hazard ratios (HR):

HR > 1: Higher hazard

HR < 1: Lower hazard

Risk between groups is constant over time (PH assumption).

Assumptions:

1. Interaction effects between covariates
2. Proportional Hazards (PH)
3. Linearity
4. Multicollinearity

Assumption 1: Interaction effects

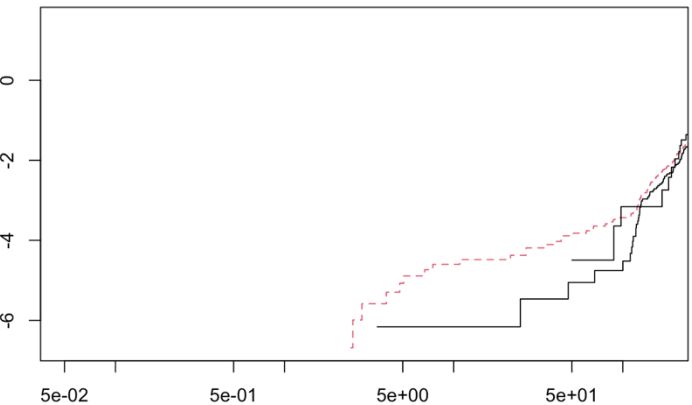
Does one variable change the effect of another?

The interaction effects of HER2 status & tumour size ($p = 0.042$) and ER status & lymph nodes examined ($p = 0.038$) were found to be significant.

Cox Proportional Hazards Ratio Assumptions

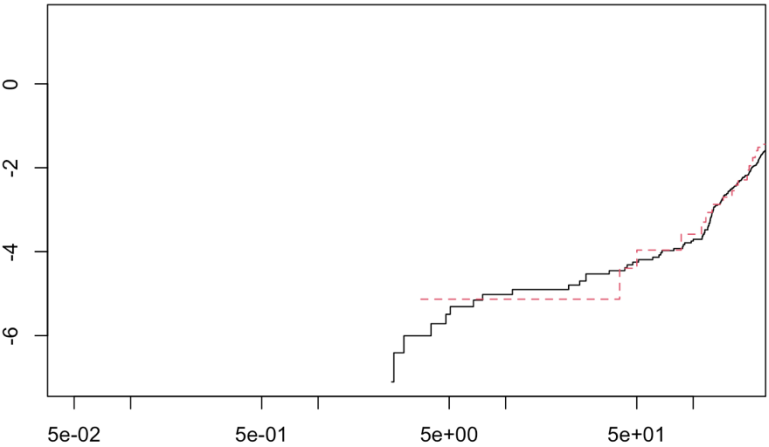
Constant Hazards Ratio

By tumour size:



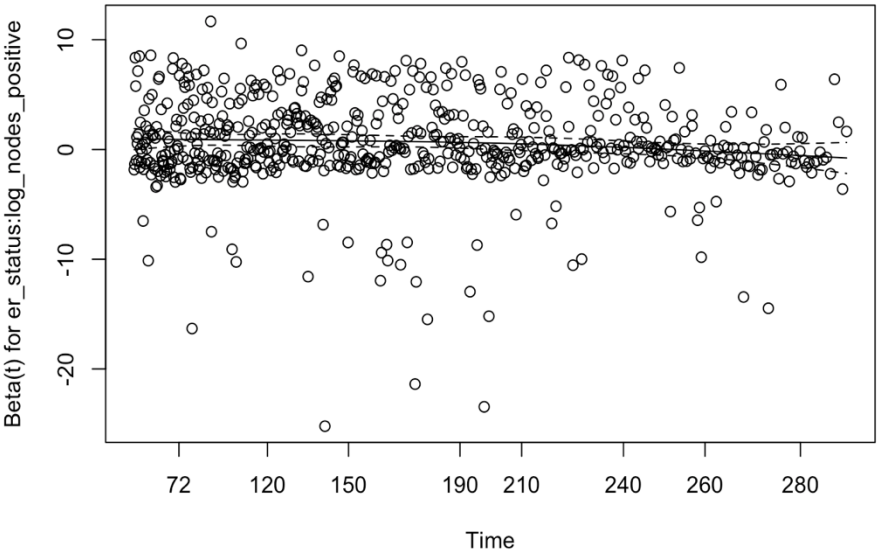
The hazard ratios between two groups should be constant (parallel).

By HER2 status:



Testing PH after fitting model:

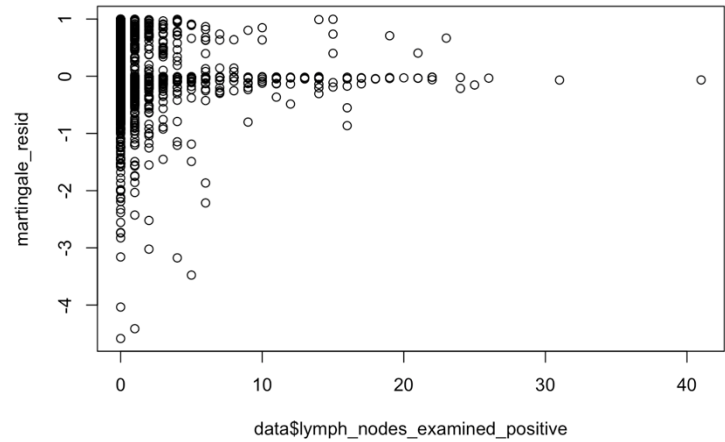
	chisq	df	p
tumor_stage	4.42021	2	0.11
age_at_diagnosis	0.00107	1	0.97
her2_status	0.42779	1	0.51
log_tumor_size	0.30456	1	0.58
er_status	0.39757	1	0.53
log_nodes_positive	0.03101	1	0.86
her2_status:log_tumor_size	0.27217	1	0.60
er_status:log_nodes_positive	1.05799	1	0.30
GLOBAL	14.54107	9	0.10



Linearity assumption

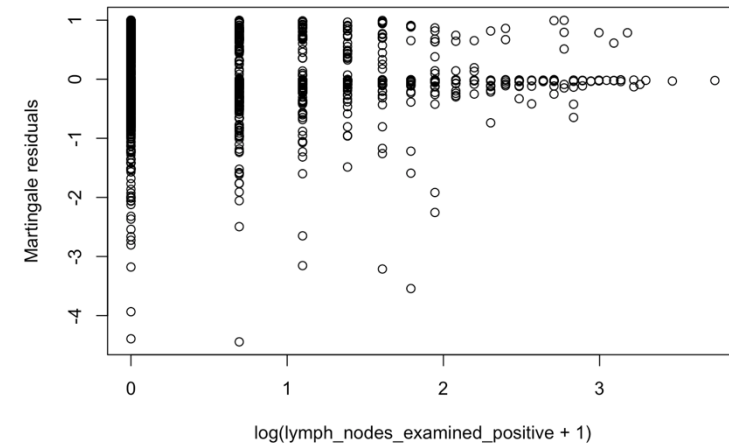
Before log transformation

Lymph nodes examined positive

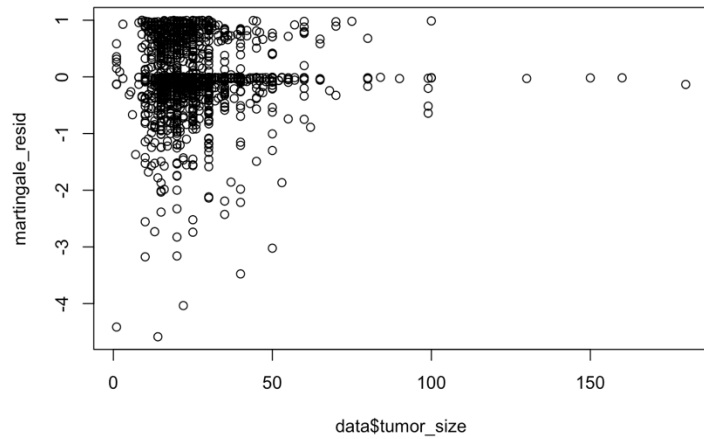


After log transformation

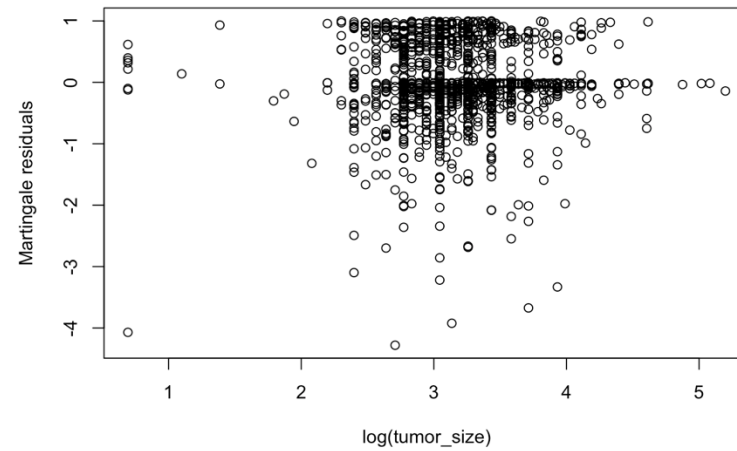
Lymph nodes examined positive



Tumour size



Tumour size



Multicollinearity test

	GVIF	Df	$GVIF^{(1/(2*Df))}$
tumor_stage	2.336190	2	1.236309
age_at_diagnosis	1.103547	1	1.050499
her2_status	17.504400	1	4.183826
log_tumor_size	1.756336	1	1.325268
er_status	1.744061	1	1.320629
log_nodes_positive	6.196117	1	2.489200
her2_status:log_tumor_size	17.275550	1	4.156387
er_status:log_nodes_positive	5.410966	1	2.326148

Cox PH Model Summary

Term	Hazards Ratio	p-value	Interpretation
tumor_stage2	1.089	0.442	Higher hazard (↑ risk)
tumor_stage3	1.302	0.324	Higher hazard (↑ risk)
age_at_diagnosis	0.988	0.001**	Lower hazard (↓ risk)
her2_statusPositive	0.402	0.103	Lower hazard (↓ risk)
log_tumor_size	0.906	0.340	Lower hazard (↓ risk)
er_statusPositive	0.737	0.018*	Lower hazard (↓ risk)
log_nodes_positive	0.718	0.036*	Lower hazard (↓ risk)
her2_status_positive:log_tumor_size	1.472	0.034*	Higher hazard (↑ risk)
er_statusPositive:log_nodes_positive	1.768	0.0004***	Higher hazard (↑ risk)

Tumor stage (2 and 3)-not a significant covariate.

Age at diagnosis showed a small but significant association: older age → slightly lower hazard.

ER-positive status is protective, associated with lower hazard.

Larger tumours increase hazard more in HER2+patients.

More positive nodes greatly increase hazard in ER+ patients.

Takeaway

Clinically, these results suggest that lymph node burden modifies the prognostic effect of ER status, and tumor size modifies the effect of HER2 status. This highlights the importance of considering molecular markers jointly with tumor burden rather than in isolation.

Limitations

Recommendations

Omitted variable bias

Cancer survival is very complex. HER2 status may be affected by other tumour characteristics such as histologic subtype and hormone receptor status.



Adding other covariates

Add more interaction effects into the model and use transformations to fit a better model with more predictive power.

Proportional hazards assumption

This assumption may still be violated if different treatments have varying time-related effects.



Observing the effect of time

- Plot the hazards in several different ways to observe the effect of time on the covariates properly
- Incorporate time-varying effects (accelerated failure time model) rather than the standard Cox PH model.

Future outlook

Research Aim: We identified the significant covariates of survival probability of breast cancer patients, which could **help future clinical diagnosis to help doctors tailor treatment plans to a patient's individual risk profile.**

Future Outlook #2: Influence of Treatments

Further research could examine **how survival outcomes change when specific treatments are applied to subgroups** (e.g. HER2+ with targeted hormone therapy).

Future Outlook #1: Predictive Tool

A future direction could involve **translating our statistical findings into a practical** tool, such as a survival prediction calculator for clinicians or patients.